

Fig. 2: An actual-size, single-side PCB for the temperature monitor for electronic equipment

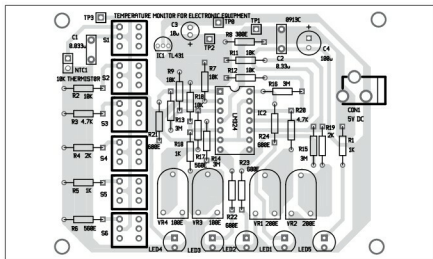


Fig. 3: Component layout for the PCB

with preset VR1. Switches S2 through S6 are used to simulate voltage levels corresponding to different temperatures.

A4 is tuned for 45°C by keeping switch S3 in closed position and trimming preset VR1 until LED1 glows. S1 should be open during calibration. The same procedure is repeated for comparator A3, A2 and A1 to tune them for 65°C, 85°C and 105°C. Table I shows the presets and LEDs corresponding to each temperature level. After calibration, open switches S2 through S6.

Construction and testing

An actual-size, single-side PCB for the temperature monitor is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable box. Fix connector CON1 for the power supply.

To test the circuit for proper functioning, verify input power supply at TP1 with respect to TP0. Check 2.5V reference voltage at TP2. The voltage at TP3 will vary with the temperature. ■

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